



Vaidarbhi Buddhe

Integrated M.Tech (Chemical Engineering) – Minor: Petroleum & Petrochemical Engineering

Institute of Chemical Technology, Mumbai, Marathawada Campus, Jalna
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CAREER OBJECTIVE

Aspiring Chemical Engineer with strong industrial and research experience, seeking to contribute to process engineering, process optimisation, data analysis, and sustainable manufacturing. Eager to apply analytical skills, simulation tools, and engineering knowledge to enhance process safety, operational efficiency, energy performance, and sustainability across the chemical and petrochemical industry.

EDUCATION

Institute of Chemical Technology, MARJ Jalna	<i>2022 – Expected 2027</i>
Integrated M.Tech in Chemical Engineering (Major) – Minor: Petroleum and Petrochemical Engineering Overall CGPA: 8.73 (Class Rank: 3/45)	
J M Patel Arts, Commerce and Science College, Bhandara	<i>2021 – 2022</i>
H.S.C. – 88.17%	
Nutan Kanya High School, Bhandara	<i>2019 – 2020</i>
S.S.C. – 100%	

RESEARCH / INDUSTRIAL EXPERIENCE

CSIR – Indian Institute of Petroleum (IIP), Dehradun (UK)	<i>Nov 2025 – Feb 2026</i>
• Project: “Thermodynamic and Experimental Analysis for Syngas to LPG Conversion through Modified Route.” • Prepared Cu–ZnO–ZrO₂–Al₂O₃ (CZZA) catalysts using the co-precipitation method, involving controlled pH and temperature, followed by drying and calcination. • Evaluated catalyst/product samples using Gas Chromatography (GC) as part of the experimental investigation of syngas-to-LPG conversion. • Performed thermodynamic analysis of a 27-reaction network and simulated the process using Aspen Plus (RGibbs), conducting sensitivity analysis across temperature, pressure and CO/H₂ ratio to evaluate LPG formation.	
Ultra Plus Lubes Pvt. Ltd., Panvel (MH)	<i>April 2025 – Jun 2025</i>
• Supervised a 12-member production team across lubricant blending, filling, packing and dispatch operations, with exposure to both PLC-controlled and manual blending systems. • Conducted daily production audits, documentation and data collection to analyse productive time, identify operational losses, and track oil losses across stages from blending to filling. • Monitored daily oil wastage and process performance, tracing major loss points across the production line to support improved operational efficiency and production control.	
Indorama Synthetics (IND) Ltd., Nagpur (MH)	<i>Aug 2024 – Oct 2024</i>
• Project: “Enhancing Energy Efficiency in CF-HTM Systems Through Insulation Upgrades.” • Analysed the impact of a proposed insulation upgrade from mineral wool to calcium silicate, assessing its effect on surface temperature, heat loss, energy consumption and emissions. • Supported the assessment of reducing surface temperature from 6°C to 3°C and achieving a 50% reduction in heat loss along with associated energy and emissions savings. • Participated in Kaizen activities to identify and implement continuous, incremental process improvements on the plant floor and completed Lean Six Sigma Green Belt training alongside plant employees.	
Sanjay Techno Products, Aurangabad (MH)	<i>Feb 2024 – Mar 2024</i>
• Studied injection moulding operations for engineering and commodity polymers, analysing defects such as flash, sink marks, burn marks, voids, jetting and warpage.	

Clarion Organics Ltd., Tumsar (MH)

Aug 2023 – Sep 2023

- Studied industrial manufacturing processes for **pharmaceutical intermediates**, including nitration, reduction and hydrogenation, along with downstream filtration, centrifugation and drying operations.
- Performed **Karl Fischer titration and factor determination** for moisture analysis, gaining hands-on exposure to laboratory testing and analytical workflows.
- Observed and studied process equipment including **glass-lined and SS316 reactors, hydrogenators, centrifuges, sparkler filters and dryers** used in pharmaceutical-intermediate production.

CERTIFICATIONS & AWARDS

Six Sigma White Belt	Jul 2026
Aspen Plus – Getting Started, ChemEngGuy	Aug 2025
Introduction to Symbolic Math with MATLAB	Aug 2025
Vedvalli Vaidyanathan Girl Student Award, ICT	2022–2023

POSITIONS OF RESPONSIBILITY

Off-Campus Representative – The Bombay Technologist, ICT MARJ	2025–Present
• Represent the institute externally by coordinating outreach initiatives and fostering engagement with students.	
Deputy Head – Literature Club, ICT MARJ	2024–2025
• Lead the planning and execution of literary events, competitions, and club activities while mentoring members and promoting creative expression.	
Member – Margjal Editorial Board	2022–2024
• Contribute to content creation, editing, and publication of the institute's editorial magazine, ensuring high-quality and engaging material.	
Event and Competition Team Member – Aakriti (Tech Fest)	2023–2024
• Assisted in organising technical events and competitions by coordinating logistics, participant engagement, and smooth event execution.	
Science Outreach Volunteer, Rainbow EduFest, Rotary club of Jalna	Feb 2023
• Contributed to a Rotary Club science outreach initiative that engaged 500+ students across 15 schools through interactive science demonstrations and awareness activities.	

EXTRA-CURRICULAR

2nd Place, Government of Maharashtra State Level Rugby Competition	2019–2020
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TECHNICAL KNOWLEDGE

Process Simulation: Aspen Plus, Aspen HYSYS, DWSIM

Chemical Engineering & Chemical Processing: Process Optimisation, Material & Energy Balances, Heat Transfer, Mass Transfer, Reaction Engineering, Process Safety, Catalyst Preparation

Data Analysis & Programming: Python, MATLAB, Microsoft Excel, Minitab

Laboratory & Analytical Techniques: Gas Chromatography, HPLC, Karl Fischer Titration, Viscosity Testing, Flash Point, Pour Point

Process & Energy Efficiency: Energy Efficiency Assessment, Heat-Loss Analysis, Sustainability Analysis, Continuous Improvement, Lean Six Sigma

Industrial & Engineering Tools: SAP ERP Systems, PLC, CATIA, Microsoft Word, PowerPoint

LANGUAGES & INTERESTS

Languages: English, Marathi, Hindi

Interests: Journaling, Sketching, Travelling, Learning New Languages